

Department Of Computer Science and Engineering
University Of Barishal
Numerical Methods, CSE 3107, 5th semester, Session 2022-23
(Red color highlighted names will present the corresponding lecture)

◆ **Module 1: Numerical Computing Foundations (Lectures 1–5)**

Lecture 1: Introduction to Numerical Computing → 230102002, HUMAYRA ALAMGIR NUHA

- Role in CSE and engineering
- Mathematical modeling pipeline
- Algorithm vs analytical solution
- Case studies (ML, graphics, simulations)

Lecture 2: Floating-Point Arithmetic → 230102024, SHOUVIK BISWAS

- Binary representation
- IEEE standard
- Rounding error
- Machine epsilon

Lecture 3: Error Analysis → 230102028, RAUFE HASAN RUHAN

- Absolute/relative error
- Truncation vs round-off
- Error propagation
- Catastrophic cancellation

Lecture 4: Conditioning and Stability → 230102013, PARTHA SARATHI BISWAS

- Well vs ill-conditioned problems
- Condition number
- Backward vs forward error

Lecture 5: Taylor Series & Order of Accuracy → 230102016, MD. TANVIR AHMED RANA

- Taylor expansion
 - Big-O notation
 - Local vs global error
 - Basis for numerical schemes
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◆ **Module 2: Numerical Linear Algebra (Core for CSE) (Lectures 6–14)**

Lecture 6: Linear Systems Overview → 230102018, DEEP PAUL

- Matrix norms
- Condition number
- Computational complexity

Lecture 7: Gaussian Elimination → 230102003, DIPANNITA GAIN PUJA

- Algorithm
- Operation count
- Numerical issues

Lecture 8: Pivoting and Stability → 230102005, NABHAN ABDULLAH

- Partial pivoting
- Complete pivoting
- Growth factor

Lecture 9: LU Decomposition → 230102021, ANTOR SARKER

- Doolittle & Crout
- Reuse for multiple RHS
- Implementation strategies

Lecture 10: Matrix Factorizations → 230102006, SARMIN SULTANA

- Cholesky decomposition
- When to use it
- Symmetric positive definite matrices

Lecture 11: Iterative Methods I – Jacobi & Gauss-Seidel → 230102022, MD. YEASIN

- Convergence conditions
- Diagonal dominance
- Spectral radius

Lecture 12: Iterative Methods II – SOR → 230102023, ABDULLAH AL MARUF

- Relaxation parameter
- Convergence acceleration

Lecture 13: Sparse Matrices & Large-Scale Systems → 230102007, S.M. SHAHRIAR SOUROV

- Storage schemes
- CSR format
- Applications in PDEs & ML

Lecture 14: Least Squares & Overdetermined Systems → 230102008, MD. SOJIB MIA

- Normal equations
 - QR decomposition (conceptual)
 - Geometric interpretation
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◆ Module 3: Nonlinear Equations (Lectures 15–21)

Lecture 15: Root-Finding Problem → 230102029, MD. REZOAN HOSSAIN

- Bracketing vs open methods
- Convergence order

Lecture 16: Bisection & False Position → 230102030, NILOY DEB

- Convergence proof
- Error bounds

Lecture 17: Fixed-Point Iteration → 230102034, MAHBUBUR RAHMAN ANTOR

- Contraction mapping
- Convergence condition

Lecture 18: Newton-Raphson Method → 230102032, SADIA MOSTARIN SOUVRA

- Derivation
- Quadratic convergence
- Failure cases

Lecture 19: Secant Method → 230102039, MD. SHIHAB

- Superlinear convergence
- Practical comparison

Lecture 20: Systems of Nonlinear Equations → 230102038, DIP DEY

- Multivariable Newton
- Jacobian matrix

Lecture 21: Nonlinear Least Squares (CSE-Oriented) → 230102037, MD. JUBAYER HASAN

- Gauss–Newton method
 - Brief intro to Levenberg–Marquardt
 - Applications in ML & data fitting
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◆ **Module 4: Interpolation & Approximation (Lectures 22–28)**

Lecture 22: Interpolation Fundamentals → 230102026, MOHAMMOD ABDULLAH AROF SADIM

- Interpolation vs approximation
- Error formula

Lecture 23: Lagrange Interpolation → 230102025, MD. AHASANUL ISLAM

- Polynomial construction
- Runge phenomenon

Lecture 24: Newton's Divided Differences → 230102019, MD. MAHFUJ ABDULLAH

- Different difference operators
- Incremental computation
- Practical advantages

Lecture 25: Spline Interpolation → 230102027, M.N. ISTIAQ

- Linear spline
- Cubic spline
- Smoothness conditions

Lecture 26: Least Squares Approximation → 230102040, MOUMITA GHOSE

- Linear regression
- Polynomial regression
- Conditioning issues

Lecture 27: Exponential & Trigonometric Fitting → 230102047, ABEA AMRIN LINAT

- Log transformation
- Linearization
- Signal processing relevance

Lecture 28: Orthogonal Basis & Fourier Approximation (Intro) → 230102012, ABDUR RAHMAN

- Trigonometric approximation
 - Connection to signal processing
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◆ Module 5: Numerical Differentiation & Integration (Lectures 29–33)

Lecture 29: Numerical Differentiation → 230102011, SYEDA FAZILATUNNASSA

- Forward/backward/central difference
- Error comparison

Lecture 30: Higher-Order Derivatives → 22CSE014, EMON RAYHAN

- Second derivative schemes
- Stability considerations

Lecture 31: Newton-Cotes Integration → 230102015, MD. READUL HAQUE SOHAN

- Trapezoidal rule
- Simpson's rules
- Composite forms

Lecture 32: Gaussian Quadrature → 230102048, MD. SUMON

- Orthogonal polynomials
- Gauss-Legendre

Lecture 33: Romberg Integration → 230102031, SAJAL TIKADAR

- Richardson extrapolation
 - Accuracy improvement
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◆ Module 6: Ordinary Differential Equations (Lectures 34–39)

Lecture 34: ODE Basics → 230102020, ANUSUA BISWAS NIDHI

- Initial value problems
- Existence & uniqueness
- Picard iteration (conceptual)

Lecture 35: Taylor Series Method → 230102035, MD SHAHIN ALOM

- Higher-order expansion
- Local truncation error

Lecture 36: Euler & Modified Euler → 230102043, RITU AKTER SAMIA

- Global error
- Stability issues

Lecture 37: Runge-Kutta Methods → 230102046, MD. AZAD

- RK2 & RK4
- Order comparison

Lecture 38: Multistep Methods → 230102042, TASNIM AHMED

- Adams-Bashforth
- Adams-Moulton
- Milne's predictor-corrector

Lecture 39: Stability & Stiff Equations → 22CSE021, MAIMUNA AKTER

- Stability region
 - Introduction to stiff systems
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◆ Module 7: Partial Differential Equations (Lectures 40–42)

Lecture 40: Classification of PDEs → 230102044, MD. OMAR FARUK

- Elliptic, parabolic, hyperbolic
- Boundary conditions

Lecture 41: Finite Difference for Heat Equation → 230102050, SANJIDA AKTER MIM

- Explicit method
- Implicit method
- Stability (CFL condition)

Lecture 42: Wave & Laplace Equations → 230102051, SABIHA AKTER

- Discretization
 - Sparse system formulation
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◆ Module 8: Numerical Optimization & Modern Computing (Lectures 43–44)

Lecture 43: Numerical Optimization → 22CSE047, ARAFAT RAHAMAN

- Golden section search
- Gradient descent
- Newton's method for optimization

Lecture 44: Modern Scientific Computing in CSE → 230102053, AFSANA

- Numerical libraries (NumPy, SciPy concepts)
- Computational complexity
- Applications in ML & AI